Advanced Trigonometry

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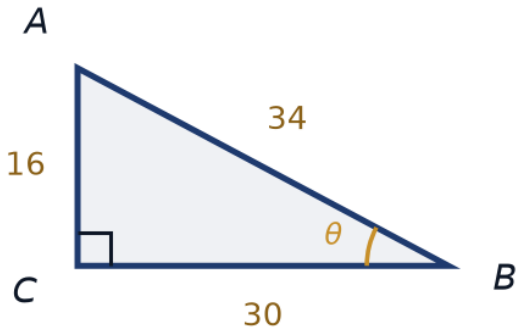
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Compound angle formulas

Expand $\sin(x + 30^\circ)$ in terms of $\sin x$ and $\cos x$, using exact values for $\sin 30^\circ$ and $\cos 30^\circ$.

2 [Foundational] • Target: Compound angle formulas

Given $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, with A and B both acute, find $\cos(A + B)$.



3 [Foundational] • Target: Compound angle formulas

Use the compound-angle formula for tangent to find the exact value of $\tan 75^\circ$, using $75^\circ = 45^\circ + 30^\circ$.

4 [Foundational] • Target: Compound angle formulas

Use the compound-angle formula for cosine to find the exact value of $\cos 15^\circ$, using $15^\circ = 45^\circ - 30^\circ$.

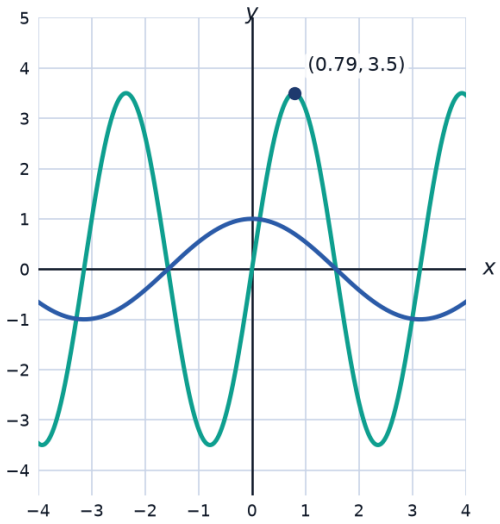
5 [Foundational] • Target: Double angle formulas

Given $\sin\theta = \frac{4}{5}$ with θ acute, find $\sin 2\theta$ and $\cos 2\theta$.

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6 [Foundational] • Target: Double angle formulas

Given $\cos\theta = -\frac{1}{3}$ with $90^\circ < \theta < 180^\circ$, find $\cos 2\theta$ and state which quadrant 2θ lies in.



7 [Foundational] • Target: Double angle formulas

Given $\tan\theta = \frac{1}{2}$ with θ acute, find $\tan 2\theta$, using $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The R sin(θ+α) form

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$, with $R > 0$ and $0^\circ < \alpha < 90^\circ$, giving α to the nearest degree.

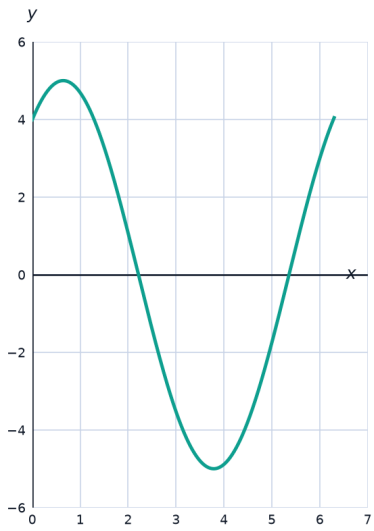
9 [Intermediate] • Target: The R sin(θ+α) form

Express $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$, giving exact values of R and α.

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10 [Intermediate] • Target: Using the R-form to find extrema and solve equations

The graph shows $y = 5\sin(x + 53^\circ)$, the R-form of $3\sin x + 4\cos x$ found earlier. State the maximum and minimum values of $3\sin\theta + 4\cos\theta$, and the value of θ (to the nearest degree, $0^\circ \le \theta < 360^\circ$) at which the maximum occurs.



11 [Intermediate] • Target: Using the R-form to find extrema and solve equations

Solve $\sqrt{3}\sin\theta - \cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using the R-form $2\sin(\theta - 30^\circ)$ found earlier.

12 [Intermediate] • Target: Solving equations that combine several identities

Solve $2\cos 2\theta + 3\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$, giving answers to one decimal place where needed.

13 [Intermediate] • Target: A well-designed word problem: alternating current

This question has three parts. The voltage in an AC circuit is modeled by $V(t) = 6\sin(100\pi t) + 8\cos(100\pi t)$ volts, where t is time in seconds. (a) Express $V(t)$ in the form $R\sin(100\pi t + \alpha)$, giving R exactly and α in radians to four decimal places. (b) State the maximum voltage reached, and the smallest positive value of t (in seconds, to four significant figures) at which it occurs. (c) Find the two values of t in $0 \le t < 0.02$ at which $V(t) = 5$, to four significant figures.

14 [Intermediate] • Target: Proving trigonometric identities

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 4.01 Differentiation

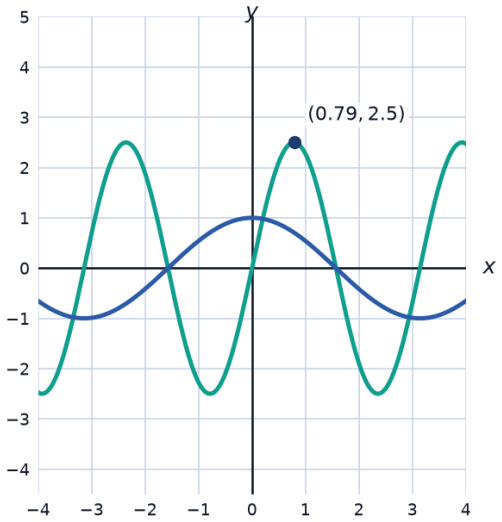
Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

16 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 5.01 Integration

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

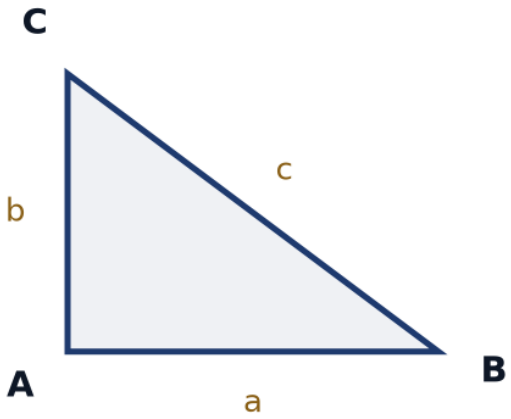
17 [Challenging] • Target: Compound angle formulas • Cross-Lesson: 6.01 Sequences & Series

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \leq \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.



18 [Challenging] • Target: Double angle formulas • Cross-Lesson: 7.01 Vectors

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



19 [Challenging] • Target: Solving equations using double angle formulas • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

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20 **[Challenging]** • **Target: The R sin(θ+α) form** • **Cross-Lesson: 9.01 Complex Numbers**

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Compound angle formulas**

Express $5\sin\theta + 12\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 7

b. 13

c. 17

d. 14

22 **[Foundational]** • **Target: Double angle formulas**

Solve $\sin(\theta) = \frac{\sqrt{2}}{2}$ for $\theta \in [0^\circ, 360^\circ)$. How many solutions are there?

a. 2

b. 3

c. 1

d. 4

23 **[Foundational]** • **Target: Solving equations using double angle formulas**

Find the maximum value of $8\sin\theta + 15\cos\theta$ as θ varies over all real values.

a. 17

b. 23

c. 15

d. 8

24 **[Foundational]** • **Target: The R sin(θ+α) form**

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

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25 [Foundational] • Target: Using the R-form to find extrema and solve equations

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)

a. $6 - 24$

b. 22

c. $2 - 64$

d. $6 + 24$

26 [Foundational] • Target: Proving trigonometric identities

Find the exact value of $\cos 75^\circ$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$

b. $\frac{-\sqrt{2}+\sqrt{6}}{4}$

c. $\frac{\sqrt{6}+\sqrt{2}}{4}$

d. $\frac{-\sqrt{6}+\sqrt{2}}{4}$

27 [Foundational] • Target: Solving equations that combine several identities

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

28 [Foundational] • Target: A well-designed word problem: alternating current

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Compound angle formulas

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

30 [Intermediate] • Target: Double angle formulas

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

31 [Intermediate] • Target: Solving equations using double angle formulas

Using the compound angle formula, expand $\sin(x + 90^\circ)$.

a. $\cos x$

b. $-\cos x$

c. $\sin x$

d. $-\sin x$

32 [Intermediate] • Target: The R sin(θ+α) form

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$

d. $\frac{7}{10}$

33 [Intermediate] • Target: Using the R-form to find extrema and solve equations

If $\sin\theta = \frac{3}{5}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{37}{50}$

b. $\frac{16}{25}$

c. $\frac{4}{5}$

d. $\frac{9}{25}$

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34 [Intermediate] • Target: Proving trigonometric identities

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2 - 64$

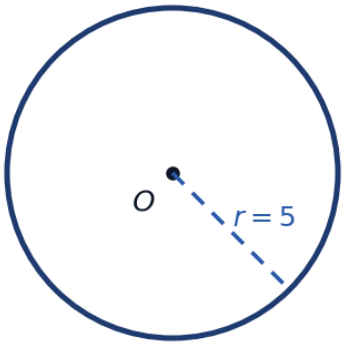
b. $\frac{1}{2}$

c. 22

d. $6 - 24$

35 [Intermediate] • Target: Solving equations that combine several identities

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

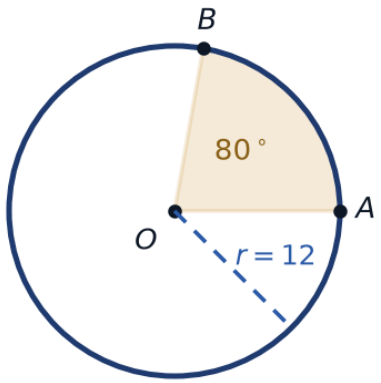
c. 1

d. $\frac{\sqrt{2}}{2}$

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36 **[Intermediate]** • **Target: A well-designed word problem: alternating current**

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

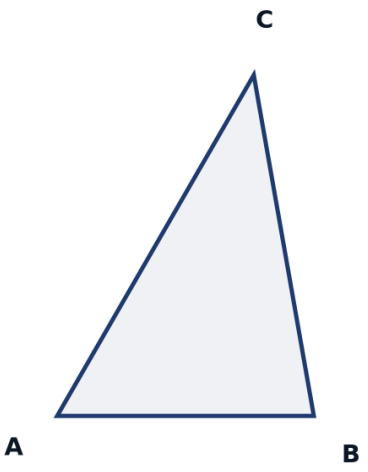
b. $\sin\theta$

c. $\tan\theta$

d. 1

37 **[Intermediate]** • **Target: Compound angle formulas**

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.



a. 5.7

b. 6.7

c. 7.7

d. 4.7

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Double angle formulas** • **Cross-Lesson: 7.01 Vectors**

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD.

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

39 **[Challenging]** • **Target: Solving equations using double angle formulas** • **Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $\frac{4}{5}$

40 **[Challenging]** • **Target: The R sin(θ+α) form** • **Cross-Lesson: 9.01 Complex Numbers**

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

41 **[Challenging]** • **Target: Using the R-form to find extrema and solve equations** • **Cross-Lesson: 1.01 Functions: Modulus & Inverse**

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. 41.4°

b. 46.4°

c. 138.6°

d. 36.4°

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42 [Challenging] • Target: Proving trigonometric identities • Cross-Lesson: 10.01 Numerical Methods

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4 °

b. 135.6 °

c. 44.4 °

d. 39.4 °

43 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 11.01 Differential Equations

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120°$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. 14.8

c. 15.8

d. 12.2

44 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 12.01 Poisson Distribution

Use the sum formula to find $\sin(30° + 60°)$.

a. $\frac{11}{10}$

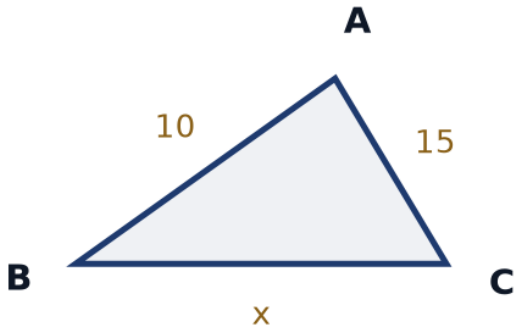
b. $\frac{1}{2} + \frac{\sqrt{3}}{2}$

c. -1

d. 1

45 [Challenging] • Target: Compound angle formulas • Cross-Lesson: 13.01 Continuous Random Variables

In triangle ABC, angle ABC = θ degrees, and angle ACB = 2θ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 cm^2, find the value of x, to 3 significant figures.



a. 13.7

b. 11.4

c. 12.8

d. 14.9

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